

C225

Research Questions and Literature Review

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A middle-school team in a building that houses 700 students sees chronic absenteeism rise after a switch to block scheduling. Their search finds no studies of block schedules in rural middle schools. Which statement is the research problem?

- A. The purpose of this study is to examine block scheduling and attendance.
- B. What is the relationship between block scheduling and attendance among rural middle school students?
- C. Absenteeism has risen after the move to block scheduling, and little research examines block schedules in rural middle schools.
- D. Block scheduling causes absenteeism in all middle schools.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A graduate student who teaches at a charter school with a strong arts program drafts four research questions. Which is most researchable?

- A. Should schools require independent reading?
- B. What is the relationship between weekly independent-reading minutes and reading fluency scores among fourth graders at two schools?
- C. Does independent reading always improve every student's reading?
- D. Is reading important?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Working late because her proposal is due Friday, a student finds two sources: a journal article in which the authors report their own study of peer tutoring, and a textbook chapter that summarizes ten tutoring studies. Which is the primary source?

- A. Neither; only raw data files count as primary.
- B. Both, since both are published.
- C. The journal article reporting the authors' own study.
- D. The textbook chapter, because it covers more studies.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A researcher wants a secondary source for an overview before reading original studies. Which fits?

- A.** A published review article summarizing past studies on the topic.
- B.** A teacher's blog post sharing her opinion.
- C.** Raw survey data from the researcher's own school.
- D.** An article in which the author reports her original experiment.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

While searching a website that ranks well in search results, a student finds four sources on homework and achievement. Which should anchor the literature review?

- A.** A white paper from a company selling a homework app.
- B.** A peer-reviewed journal article that reports its sample, methods, and results.
- C.** Whichever source ranks first, since search ranking reflects quality.
- D.** A popular magazine article quoting teachers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A student drafted her literature review over a spring break at the beach. It discusses Study A, then Study B, then Study C, each in its own paragraph. Her advisor says it summarizes but does not synthesize. What is the best revision?

- A.** Reorganize by themes, comparing and contrasting findings across studies and pointing out the gap.
- B.** Remove studies that disagree with the others.
- C.** Add more studies in the same one-per-paragraph format.
- D.** Put the studies in order by publication year.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A researcher at a university with 30,000 students predicts that students who use spaced practice will score higher on a final exam than students who cram. What is the null hypothesis?

- A. Spaced-practice students will score higher than cramming students.
- B. Spaced practice always produces higher scores.
- C. Cramming students will score higher.
- D. There is no difference in exam scores between spaced-practice and cramming students.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

“Students in the tutoring group will have higher reading scores than students in the control group.” How is this statement classified?

- A. A nondirectional alternative hypothesis
- B. A null hypothesis
- C. A directional alternative hypothesis
- D. A research question

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

In a district that recently gave laptops to every student, researchers compare math-anxiety scores of students taught with game-based lessons versus standard lessons. What is the dependent variable?

- A. Type of lesson
- B. The students
- C. Having a laptop
- D. Math-anxiety score

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A researcher asks whether class size affects student engagement. What is the independent variable?

- A.** Student engagement
- B.** The teacher
- C.** Both, since in education each variable always depends on the other.
- D.** Class size

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A researcher whose grant was just renewed for three years asks, “Does a new phonics program cause higher decoding scores than the current curriculum?” Which design best answers this question?

- A.** An experiment with random assignment to the new program or the current curriculum
- B.** Any design, since any data can answer any question
- C.** A correlational study of program use and scores
- D.** A survey of teacher opinions about phonics

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A researcher chose a phenomenological interview design. Which question most likely drove that choice?

- A.** What is the correlation between work hours and GPA?
- B.** How do first-generation college students describe their first semester?
- C.** What percentage of students are first-generation?
- D.** Does a mentoring program cause higher GPAs?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A middle-school team ~~in a building that houses 700 students~~ sees chronic absenteeism rise after a switch to block scheduling. Their search finds no studies of block schedules in rural middle schools. Which statement is the research problem?

Target word: identify the problem statement **Decides:** The problem names the issue and the gap.

A	Confusion set	That is a purpose statement.
B	Adjacent category	That is the research question.
C	Correct	States the issue and the gap in knowledge.
D	Absolute language	A sweeping claim, not a problem to study.

2. Answer B Concept 2

A graduate student who teaches ~~at a charter school with a strong arts program~~ drafts four research questions. Which is most researchable?

Target word: best question **Decides:** Named population and measurable variables.

A	Adjacent category	A policy or value question data cannot settle.
B	Correct	Specific population, measurable variables, answerable with data.
C	Absolute language	"Always" and "every" make it untestable.
D	Dropped element	No population or measurable variables.

3. Answer C Concept 3

~~Working late because her proposal is due Friday,~~ a student finds two sources: a journal article in which the authors report their own study of peer tutoring, and a textbook chapter that summarizes ten tutoring studies. Which is the primary source?

Target word: primary source **Decides:** Authors reporting their own study.

A	Adjacent category	In literature reviews, original study reports are primary sources.
B	Absolute language	Publication does not make a source primary.
C	Correct	It is the original report of the research.
D	Inverted operation	Summarizing others' work makes it secondary.

4. Answer A Concept 3 • transfer

A researcher wants a secondary source for an overview before reading original studies. Which fits?

Target word: reverse: find the secondary source **Decides:** A source that summarizes others' studies.

A	Correct	It interprets and summarizes others' work.
B	Adjacent category	Opinion, not a scholarly summary.
C	Confusion set	Original data, not a summary of the literature.
D	Inverted operation	That is primary.

5. Answer B Concept 4

While searching ~~a website that ranks well in search results~~, a student finds four sources on homework and achievement. Which should anchor the literature review?

Target word: strongest source **Decides:** Peer review and a transparent method.

A	Confusion set	Commercial interest; not independent evidence.
B	Correct	Scholarly and checkable.
C	Camouflage	Ranking reflects popularity, not rigor.
D	Adjacent category	Popular press is not peer-reviewed research.

6. Answer A Concept 5

A student drafted her literature review ~~over a spring break at the beach~~. It discusses Study A, then Study B, then Study C, each in its own paragraph. Her advisor says it summarizes but does not synthesize. What is the best revision?

Target word: how to synthesize **Decides:** Organize by theme and compare across studies.

A	Correct	Synthesis integrates studies around ideas.
B	Inverted operation	Contrasting findings are part of synthesis.
C	Dropped element	More summaries are still summaries.
D	Adjacent category	A chronological list is still study-by-study.

7. Answer D Concept 6

A researcher ~~at a university with 30,000 students~~ predicts that students who use spaced practice will score higher on a final exam than students who cram. What is the null hypothesis?

Target word: state the null **Decides:** The null states no difference.

A	Inverted operation	That is the alternative (research) hypothesis.
B	Absolute language	A claim of certainty, not a testable null.
C	Confusion set	An alternative in the opposite direction, not the null.
D	Correct	The null asserts no effect.

8. Answer C Concept 6 • transfer

“Students in the tutoring group will have higher reading scores than students in the control group.” How is this statement classified?

Target word: classify the hypothesis **Decides:** Predicts an effect and names its direction.

A	Dropped element	Drops the word “higher,” which gives direction.
B	Inverted operation	The null states no difference.
C	Correct	Predicts a difference and which group is higher.
D	Adjacent category	It is a prediction, not a question.

9. Answer D Concept 7

In a district ~~that recently gave laptops to every student~~, researchers compare math-anxiety scores of students taught with game-based lessons versus standard lessons. What is the dependent variable?

Target word: dependent variable **Decides:** The outcome measured is math anxiety.

A	Inverted operation	That is the independent variable.
B	Confusion set	Participants, not a variable.
C	Camouflage	Every student has one; it does not vary.
D	Correct	The outcome that may depend on lesson type.

10. Answer D Concept 7 • transfer

A researcher asks whether class size affects student engagement. What is the independent variable?

Target word: independent variable **Decides:** Class size is the presumed cause.

A	Inverted operation	That is the outcome, the dependent variable.
B	Confusion set	Not the variable being studied.
C	Absolute language	The question sets a direction: size affecting engagement.
D	Correct	The factor thought to influence the outcome.

11. Answer A Concept 8

A researcher ~~whose grant was just renewed for three years~~ asks, “Does a new phonics program cause higher decoding scores than the current curriculum?” Which design best answers this question?

Target word: match design to a cause question **Decides:** “Cause” needs manipulation and comparison.

A	Correct	Only an experiment supports a causal claim.
B	Absolute language	Question type limits which designs fit.
C	Adjacent category	Shows association, not cause.
D	Confusion set	Measures beliefs, not decoding outcomes.

12. Answer B Concept 8 • transfer

A researcher chose a phenomenological interview design. Which question most likely drove that choice?

Target word: reverse: from design to question **Decides:** Phenomenology studies lived experience.

A	Adjacent category	A relationship question needs a correlational design.
B	Correct	Asks about lived experience.
C	Confusion set	A descriptive quantitative question.
D	Inverted operation	A cause question needs an experiment.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Problem, purpose, question	The research problem names the issue and the gap; the purpose states what the study will do; the research question asks something specific the data can answer.	Mistaking a research question or purpose statement for the problem.	“Which statement is the research problem?”
2	Researchable questions	A researchable question names a population and variables or phenomena that can be observed or measured, and avoids value judgments like “should.”	Accepting a broad, yes/no, or policy question.	Several candidate questions, one specific and measurable.
3	Primary vs. secondary sources	A primary source is the original report by the people who did the study; a secondary source summarizes or interprets others’ work.	Thinking broader coverage makes a source primary.	A journal study vs. a textbook or review.
4	Scholarly quality of sources	Core evidence should come from peer-reviewed, methodologically transparent research; popular, commercial, and opinion pieces are weaker.	Trusting search ranking, popularity, or a vendor’s claims.	A list of sources of mixed types.
5	Synthesis vs. summary	A literature review synthesizes: it organizes by theme, compares and contrasts findings, and shows the gap the study fills.	Writing one paragraph per study.	“Summarizes but does not synthesize.”
6	Null and alternative hypotheses	The null states no difference or no relationship; the alternative states the expected effect, directional if it names which group is higher.	Writing the prediction as the null.	A predicted difference between groups.
7	Independent vs. dependent variables	The independent variable is the presumed cause or grouping; the dependent variable is the outcome measured.	Swapping them, or naming the participants or setting as a variable.	“effect of X on Y,” “compare Y across groups of X.”
8	Question type drives design	Cause questions need experimental designs; relationship questions need correlational designs; lived-experience questions need qualitative designs.	Answering a cause question with a correlational or opinion design.	Verbs like “cause,” “relate to,” or “describe the experience of.”

Course scope (public description): The Research Questions and Literature Reviews course focuses on how to conduct a thorough literature review that addresses and identifies important educational research topics, problems, and questions, and helps determine the appropriate kind of research and data needed to answer one's research questions and hypotheses. Research Foundations is a prerequisite for this course.